Echo

Data Intelligence Assistant

Model: `gemini-2.5-flash`

Supported files: CSV, XLSX

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Echo — Data Intelligence Assistant

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quarter_01_sales.csv 3.4MB

Dataset Preview

	Date	Store Location	Product Category	Product Name	Quantity Sold	Unit Price (₦)	Total Sales (₦)	Payment Method	Salesperson
0	01/01/2025	Gwarinpa	Uncategorised	Unknown Product	12	1244.11	14929.32	None	Dami
1	01/01/2025	Wuse	Household Items	Detergent	1000	486.88	486880	Transfer	Alice
2	01/01/2025	Gwarinpa	Uncategorised	Unknown Product	-1	-377.7	-500	Cash	Fatima
3	01/01/2025	Wuse	Uncategorised	Unknown Product	0	1761.54	100000	Cash	Unknown
4	01/01/2025	Wuse	Beverages	Pepsi	17	439.16	7465.72	POS	Cynthia
5	01/01/2025	Utako	Uncategorised	Unknown Product	11	1866.07	20526.77	POS	Chinedu
6	01/01/2025	Gwarinpa	Beverages	Fanta	1000	1894.04	1894040	POS	Alice
7	01/01/2025	Central Area	Snacks	Cake	19	-408.79	100000	Cash	Alice
8	01/01/2025	Gwarinpa	Household Items	Toilet Roll	0	-155.7	0	Transfer	Bola
9	01/01/2025	Wuse	Beverages	Coca-Cola	1000	265.66	265660	Transfer	Bola

Data Profile

Rows	Columns	Missing Values	Numeric Columns
49,549	9	12373	3

Automatic EDA Charts

📝 Executive Summary

✨ Generate Executive Summary

Executive Summary: Sales Data Review

This executive summary provides an overview of the provided sales dataset, highlighting key operational patterns, critical data quality issues, opportunities for improvement, and actionable recommendations.

Key Patterns & Trends

- **High Sales Activity:** The dataset covers 90 days, with a significant peak on '15/01/2025' (619 transactions). The average total sales per transaction is approximately ₦219,928.
- **Dominant Store Locations:** 'Central Area', 'Gwarinpa', and 'Wuse' are the top three store locations by transaction volume, collectively accounting for over 60% of recorded sales activities.
- **Product Categories & Names:** After the pervasive "Uncategorised" and "Unknown Product" entries, "Snacks" and "Beverages" emerge as the most frequent product categories. Popular specific products include Chin Chin, Meat Pie, Pepsi, and Fanta.
- **Payment Preferences:** POS is the most frequently recorded payment method (12,569 transactions), closely followed by Transfer and Cash.
- **Salesperson Performance:** Alice is the most active salesperson, indicating potential for benchmarking best practices.

Data Quality & Risks

- **Critical Data Inconsistencies & Errors:**
  - **Negative Values:** Presence of negative `Quantity Sold` (min -1), `Unit Price (₦)` (min -499.87), and `Total Sales (₦)` (min -500.00) indicates significant data entry errors or uncaptured business logic (e.g., returns processed as negative sales without clear distinction). **This is the most severe data quality issue, undermining the reliability of all sales metrics.**
  - **Sales Calculation Discrepancy:** The sample data clearly shows `Total Sales (₦)` is not consistently `Quantity Sold * Unit Price (₦)`. For instance, a quantity of 0 with a high total sales value, or a negative unit price leading to a large positive total sales. This fundamentally breaks the integrity of sales data.
- **Missing Critical Data:**
  - `Payment Method`: A substantial 12,372 missing values (approximately 25% of the dataset) for `Payment Method` severely limits the ability to analyze payment trends, reconciliation, and customer preferences.
  - `Quantity Sold`: One missing value for `Quantity Sold` is minor but should be addressed.
- **Poor Data Categorization:**
  - `Uncategorised` / `Unknown Product`: A staggering 19,868 entries (over 40% of the dataset) fall under "Uncategorised" product category and "Unknown Product" name. This renders granular product performance analysis nearly impossible and suggests a fundamental

flaw in product catalog management or transaction logging.

- **Inconsistent `Salesperson` Entry:** The presence of 'Unknown' as a salesperson reduces accountability and performance tracking.
- **Incorrect Data Type:** The `Date` column is stored as an `object` (string), which prevents direct temporal analysis without conversion.

Opportunities & Insights

- **Improved Operational Efficiency:** Resolving the negative values and sales calculation discrepancies offers an immediate opportunity to enhance data accuracy, improve reporting reliability, and identify potential process gaps in handling returns or corrections.
- **Enhanced Product Strategy:** Categorizing the "Uncategorised" and "Unknown Product" entries will unlock rich insights into actual product performance, inventory management needs, and opportunities for targeted marketing and promotions.
- **Deeper Payment Analysis:** Addressing missing `Payment Method` data would enable a clearer understanding of customer payment preferences, aiding in optimizing payment infrastructure and reducing transaction costs.
- **Sales Performance Optimization:** With cleaner data, detailed analysis of store, salesperson, and product performance can drive strategic decisions regarding resource allocation, sales targets, and incentive programs.
- **Temporal Trend Analysis:** Converting the `Date` column will allow for robust time-series analysis to identify seasonal trends, peak sales periods, and long-term growth or decline.

Recommendations

1. **Immediate Data Cleaning & Validation:**
   - **Prioritize Resolution of Negative Values and Calculation Errors:** Investigate the root cause of negative `Quantity Sold`, `Unit Price (₦)`, and `Total Sales (₦)`. Define clear rules for handling returns, refunds, or corrections, ensuring these are recorded accurately (e.g., as separate transaction types or with appropriate positive values).
   - **Correct Sales Calculation:** Implement a validation rule to ensure `Total Sales (₦)` is consistently and accurately calculated as `Quantity Sold * Unit Price (₦)`. Address all existing discrepancies.
   - **Convert `Date` Column:** Convert the `Date` column to a proper datetime format for time-based analysis.
2. **Product Data Enrichment:**
   - **Mandate Product Categorization:** Develop and enforce a comprehensive product categorization system. Implement processes to minimize or eliminate 'Uncategorised' and 'Unknown Product' entries moving forward.
   - **Retroactive Categorization:** Initiate a project to retroactively categorize the existing 'Uncategorised' and 'Unknown Product' entries, even if it requires manual review and mapping. This is essential for unlocking the value of historical data.
3. **Address Missing Values & Operational Gaps:**
   - **Investigate `Payment Method` Missingness:** Determine the reason for missing `Payment Method` data. Implement system changes or training to ensure this crucial field is always captured. Consider a strategy for imputing or marking historical missing values if their origin cannot be determined.
   - **Standardize Salesperson Entry:** Implement strict data entry for `Salesperson` to eliminate 'Unknown' entries, ensuring all sales can be attributed.
4. **Implement Data Governance & Quality Checks:**

- Establish ongoing data validation rules and automated checks at the point of data entry or ingestion to prevent future occurrences of negative values, missing data, and incorrect calculations.
- Provide training to data entry personnel on data quality standards and the importance of accurate record-keeping.

By addressing these critical data quality issues, the organization can transform this dataset into a reliable asset for informed decision-making, leading to improved operational efficiency and enhanced business strategies.

💬 Ask Echo

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